Hyperspectral Image Super-Resolution by Band Attention Through Adversarial Learning

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***Abstract*— Hyperspectral image (HSI) super-resolution (SR) is a challenging task due to the problems of texture blur and spectral distortion when the upscaling factor is large. To meet these two challenges, band attention through the adversarial learning method is proposed in this article. First, we put the SR process in a generative adversarial network (GAN) framework, so that the resulted high-resolution HSI can keep more texture details. Second, different from the other band-by-band SR method, the input of our method is of full bands. In order to explore the correlation of spectral bands and avoid the spectral distortion, a band attention mechanism is proposed in our generative network. A series of spatial–spectral constraints or loss functions is imposed to guide the training of our generative network so as to further alleviate spectral distortion and texture blur. The experiments on the Pavia and Cave data sets demonstrate that the proposed GAN-based SR method can yield very high-quality results, even under large upscaling factor (e.g., 8×). More importantly, it can outperform the other state-of-the-art methods by a margin which demonstrates its superiority and effectiveness.**

***Index Terms*— Adversarial learning, band attention, hyperspectral image (HSI) super-resolution (SR).**

I. Introduction

HYPERSPECTRAL sensors can capture reflectance information of materials in many contiguous and narrow spectral channels. The resulting hyperspectral image (HSI) contains rich spectral information and has been widely researched in the past decades [1]–[4]. This distinctive characteristic brings majority applications in computer vision and remote sensing fields, e.g., land cover detection, resource management, and medical diagnosis [5]–[7]. However, it is still difficult to acquire an HSI of high spatial resolution directly. This is mainly due to the limitations of hardware, such as data volume, signal-to-noise-ratio (SNR), and camera cost. Consequently, the obtained HSI is always with relatively low spatial resolution, which confines its practical applications. Therefore, HSI spatial super-resolution (SR) problem attracts much attention in the remote sensing image processing field.

Image SR is a promising signal post-processing technique aiming at acquiring a high-resolution (HR) image from a low-resolution (LR) image to overcome the inherent resolution limitations [8]. As for the RGB image SR, there have been many research studies proposed in the past decades. Recently, RGB image SR problem has achieved great progress due to the use of a convolutional neural network (CNN). Dong *et al.* [9] first proposed a CNN-based method for RGB image SR. After that, a series of novel CNN architectures [10] is proposed to improve the SR performance. All these works show that the network architecture design is the key factor affecting the performance. However, HSI SR lags far behind its RGB counterpart. More importantly, different from the RGB image, HSI consists of hundreds of spectral bands, making feature extraction of such a high-dimensional 3-D cube more difficult. It is also worth noting that, when the upscaling factor is large, texture blur and spectral distortion problems are more severe.

To handle these challenging issues, in this article, novel band attention through the adversarial learning-based SR method is proposed. First, we put the SR process in a generative adversarial network (GAN) framework. In this way, the result of our method can be of high visual quality and maintain more texture details. Second, a novel band attention mechanism is proposed, which can be embedded in our generator network (GN) and is beneficial to alleviate the spectral distortion problem. At last, a series of spatial–spectral constraints or loss functions is imposed to guide the training of GN and further alleviate spectral distortion and texture blur.

An illustration of our method is presented in Fig. 1. The main contributions of this article can be summarized as follows.

1) A novel adversarial learning-based SR method for HSI is proposed. It is totally end-to-end trainable and no need for extra priorities except an LR version of HSI. To our

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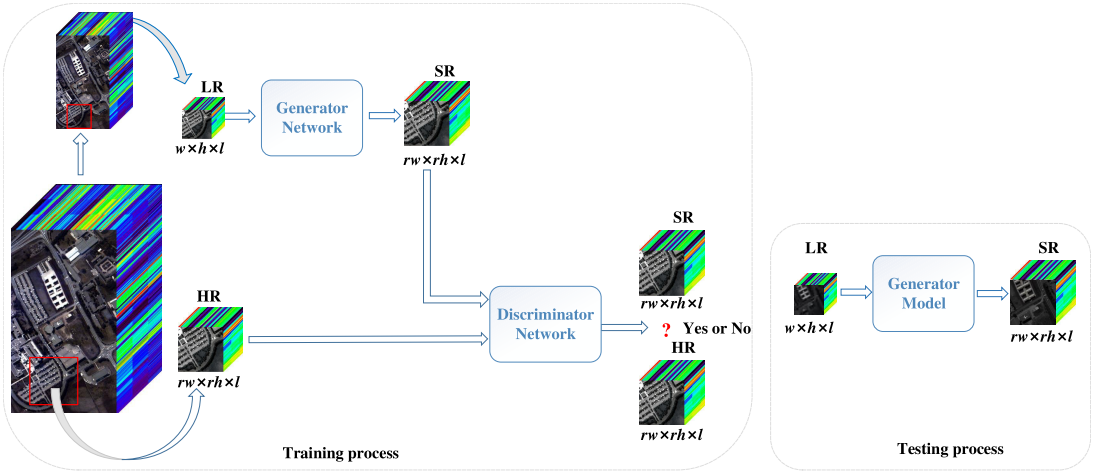


Fig. 1. Illustration of our proposed framework.

best knowledge, this is the first work which puts the HSI spatial SR problem in such an adversarial learning framework.

2) To take full consideration of the spectral distortion problem, a novel band attention mechanism is proposed, which can efficiently explore the correlation between spectral bands and benefit spectral consistency of the final result.
3) A series of novel spatial–spectral constraints or loss functions is proposed, e.g., cosine loss, adapted 3-D total variation (TV) loss, and adversarial loss. These losses are imposed on the training phase and can effectively guide the network training.

The remainder of this article is organized as follows. Section II briefly introduces SR and deep learning for HSI. Section III details our proposed SR framework based on adversarial learning and band attention with several spatial–spectral constraints. Experimental results are presented and analyzed in Section IV. The conclusions are drawn in Section V.

II. Related Works

The SR of HSI aims at restoring the original image from its downscaled version, which breaks the inherent limitation of spatial resolution in HSI. Generally, there are different approaches to enhance the spatial resolution of HSI, and two main ways are prevalent: fusion with other high spatial resolution images and single-image SR without any other prior or auxiliary information.

As for image fusion-based method, the higher spatial resolution images such as panchromatic image and multispectral image are needed as prior to assisting hyperspectral SR [11]–[15]. Traditional methods such as the Bayesian rule-based algorithm are utilized firstly [11], [12]. Then, dictionary-based fusion algorithms via spatial dictionary or spectral dictionary using sparse representation technologies are widely applied in hyperspectral SR [14]. Besides, Dong *et al.* [15] employed sparse representation technology to jointly estimate the hyperspectral dictionary and sparse codes based on spatial–spectral sparsity for enhancing spatial resolution. Hyperspectral pansharpening as a typical case of hyperspectral SR employs a corresponding panchromatic image to enhance the resolution of HSI, which has already attracted lots of attention in the literature [16]–[18]. However, in reality, higher spatial resolution images may not be easy to acquire.

Single-image-based SR does not need any other prior or auxiliary information, which has better feasibility in practice. Traditionally, the SR problem can be partially solved by filtering-based methods, such as bilinear [19] and bicubic interpolation [20]. These filtering-based methods often lead to edge blur and spectral distortions, especially when the upscaling factor is large.

Recently, deep neural network, especially the CNN, has already demonstrated its great success in computer vision [4], [21]–[24]. Inspired by these successful applications in RGB image processing, deep learning is also widely utilized in HSI. Originally, CNNs are employed in hyperspectral classification frequently for better classification accuracy [25]–[27]. Chen *et al.* [28] proved that a 3-D CNN can extract distinguished spatial–spectral features for HSI classification problem. Then, Mei *et al.* [29] showed that a 3-D CNN can also be utilized for image SR and achieved excellent performance. However, the naive 3-D CNN is not capable of capturing spectral correlation effectively and the simple MSE error cannot accurately gauge spectral distortion. Adversarial learning is a prevalent tool to address the image-generation task [30]. In particular, GAN consisted of two parts, the GN and discriminator network (DN). The GN can capture the data distribution, and the DN can estimate the probability that a sample came from the training data rather than GN. After the seed work of Goodfellow *et* al. [30], a series of modification work has been proposed. Radford *et al.* [31] proposed the DC-GANs which utilized the convolution layer instead of the fully connected layer to improve the generation ability. Arjovsky *et al.* [32] utilized the Wasserstein distance to replace the original loss function in the GAN and improve the stability of learning. Meanwhile, GAN has been utilized in a mount of application works, e.g., image style transformation [33], face generation [34], and semantic segmentation [35].

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Very recently, adversarial learning was also imposed in the RGB image SR problem. Ledig *et al.* [36] proposed an adversarial learning-based RGB image SR method, which shows that, with a powerful adversarial framework, CNNs can create plausible-looking imagery with high perceptual quality.

This article is also inspired by the great success of attention mechanism in the deep learning field. Vasmani *et al.* [37] proved that the attention mechanism is very effective in the machine translation task. Hu *et al.* [38] proposed a channel attention residual network for the image classification problem. Recently, there are many works utilizing attention mechanism to enhance their network ability for various applications works, e.g., image caption [39], person re-identification [40], and semantic segmentation [41]. Although, attention mechanism has been proved to be an effective tool for global context information fusion and local information extraction, this important mechanism has not yet been studied and utilized in the HSI SR problem.

Therefore, based on the aforementioned description and analysis, we propose novel band attention through the adversarial learning method, combining a series of spatial–spectral constraints to super-resolve the low spatial HSI.

III. Proposed Method

In hyperspectral SR, the objective is to estimate an HR super-resolved image from an LR input HSI. Here, I^{LR} is the LR version of its HR counterpart I^{HR}. In the training process, the HR version of HSI is available. I^{LR} is acquired by applying a Gaussian spatial filter to I^{HR} followed by a downsampling operation with downsampling (upsampling) factor r. Generally, the size of I^{LR} with l spectral channels is $l \times h \times w$ and I^{HR} is of size $l \times rh \times rw$.

A. 3-D GAN

Our proposed framework consists of two parts: GN and DN, and a knowledge on both is gained in the training process. The well-trained generator is employed in the testing process. The aim of the GAN-based SR framework is to train a generating function G which can create the corresponding HR for a given LR version of HSI that are highly similar to the HR image. To obtain this, we employ CNNs with several residual blocks based on band attention to train the GN. Given the training images $\mathbf{I}_i^{\mathrm{HR}}$, $i = 1{:}\ N$, and their LR counterparts $\mathbf{I}_i^{\mathrm{LR}}$, the estimated parameters θ_{G} are obtained as follows.

$$\widehat{\theta}_{\mathrm{G}} = \arg\min_{\theta_G} \frac{1}{N} \sum\nolimits_{i=1}^{N} l_{\mathrm{loss}}\left(G_{\theta_G}\left(\mathbf{I}_i^{\mathrm{LR}}\right), \mathbf{I}_i^{\mathrm{HR}}\right). \tag{1}$$

The weights estimated from $\widehat{\theta}_G$ are applied in the testing process to reconstruct the high spatial resolution HSI. The proposed SR model is to train the DN to maximize the correct recognition rate of the super-resolved images obtained from the GN and the real HSI, meanwhile to minimize the generator according to the following minimax formulation:

$$\min_{\theta_G}\max_{\theta_D} \ \mathbb{E}_{\mathbf{I}^{\mathrm{HR}}\sim p_{\mathrm{train}}(\mathbf{I}^{\mathrm{HR}})}[\log D_{\theta_D}(\mathbf{I}^{\mathrm{HR}})] + \mathbb{E}_{\mathbf{I}^{SR}\sim p_G(\mathbf{I}^{\mathrm{LR}})}[\log(1 - D_{\theta_D}(G_{\theta_G}(\mathbf{I}^{\mathrm{LR}})))]. \tag{2}$$

With the aforementioned model, the trained generator creates HR images that are highly similar to the real images, which are difficult to be recognized by the DN. This encourages that the super-resolved results have perceptually superior in the aspect of the subspace, the manifold of HSI. The detailed GN and DN of our proposed framework are depicted in Fig. 2, where different colors denote different operations. The GN is composed of three novel residual blocks with an identical layout. Each block consists of two 3-D convolutional layers with $5 \times 3 \times 3$ kernel size followed by a batch-normalization layer and a PReLU as the activation function, and the detailed information is demonstrated in Fig. 2. To increase the resolution of the input LR image, we employ transposed convolution layers. At the last layer, the tanh activation function is used.

To pursue superior generated images, the DN is trained to solve the maximization problem in (2). LeakyReLU activation is utilized to avoid max-pooling. Followed by six 3-D convolutional layers, a 3-D average pooling layer is employed to avoid overfitting. The resulting feature vectors followed by dense layers are used with a sigmoidal activation function to obtain a probability for classification.

The detailed parameter setting in the proposed architecture is illustrated in Fig. 2. Taking four times downsampling in Pavia University as an instance, the kernel size is set to $5 \times 3 \times 3$, the number of filter kernels and the stride of each layer are depicted in detail. For the CAVE data set, since the number of bands is much less than the one in the ROSIS data sets, the kernel of 3-D pooling layer is set to $4 \times 18 \times 18$.

Due to the fact that spectral distortion is a vital problem in hyperspectral SR, in our proposed framework, we propose a novel residual block structure based on band attention, then spatial-spectral-based loss functions are presented to enhance the spatial resolution while keeping the real spectral information precisely. Section III-B will introduce these two parts in detail.

B. Band Attention Mechanism in Our Network Architecture

Unlike other hyperspectral SR algorithms, our proposed framework takes the spectral bands of HSI into consideration. In order to investigate a structure that can represent the relationship between spectral bands, we design a novel residual block based on the band attention mechanism. With the goal of enhancing the quality of representation by modeling the interdependencies between the bands of its convolutional features, the proposed novel residual block must learn to employ global information to selectively emphasize informative band features and suppress less useful ones. An illustration of our proposed band attention residual block is presented in Fig. 3.

Let $\mathbf{X} \in \mathbb{R}^{c\times l\times h\times w}$ be the input of the band attention residual block, $\mathbf{B}_{\mathrm{res}} \in \mathbb{R}^{c\times l\times h\times w}$ be the output of the band attention residual block

$$\mathbf{F} = \delta\left(f_{\mathrm{conv3D}}^{2}\left(\delta\left(f_{\mathrm{conv3D}}^{1}(\mathbf{X})\right)\right)\right) \tag{3}$$

where $\mathbf{F} \in \mathbb{R}^{c\times l\times h\times w}$ denotes the current feature maps after two 3-D convolution layers $f_{\mathrm{conv3D}}^{1}(\cdot)$ and $f_{\mathrm{conv3D}}^{2}(\cdot)$, δ means the Prelu activation function.

To design a spectral band descriptor, we employ the average global 2-D pooling in its spatial dimensions $h \times w$ by using

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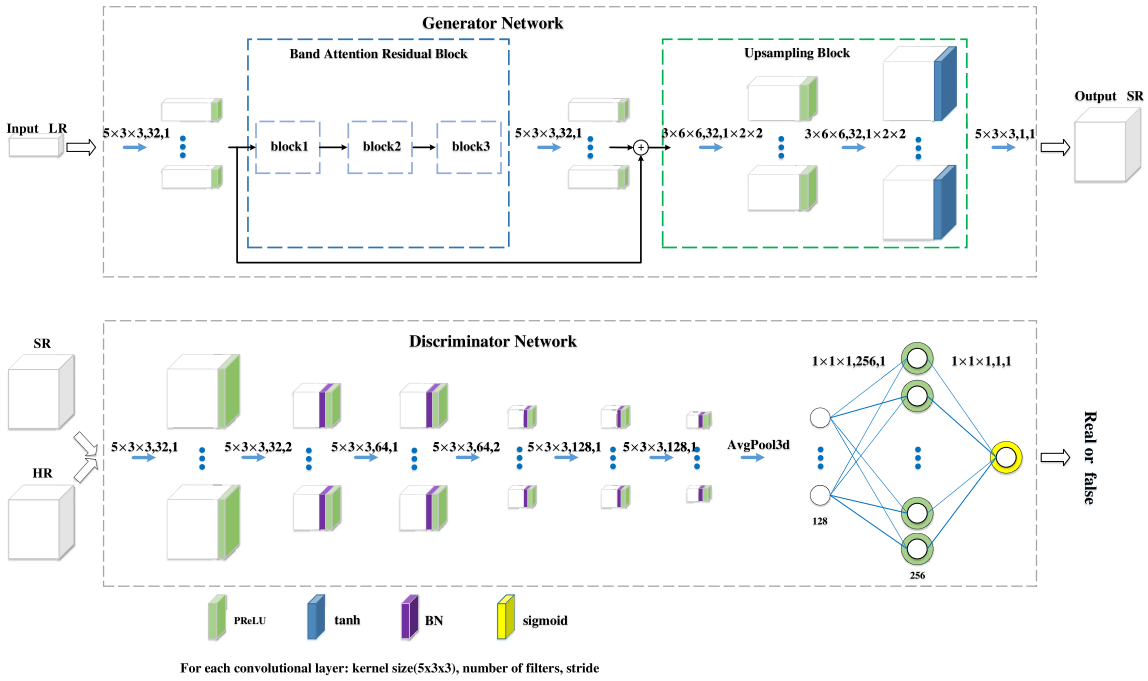


Fig. 2. Detailed architecture of GN and DN.

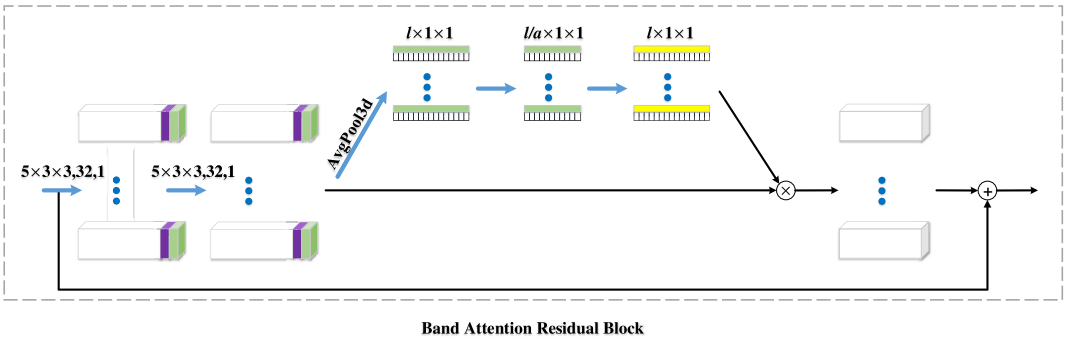


Fig. 3. Detailed structure of our band attention residual block.

the AvgPool3d function in Pytorch framework to generate bandwise statistics. In our design, bands in different channel groups are considered to have different statistics.

Firstly, a descriptor $\mathbf{s} \in \mathbb{R}^{c*l}$ is generated by global pooling $\mathbf{F}$ through its spatial dimensions $h \times w$, such that the (i, j)th elements of $\mathbf{s}$ is calculated as

$$\mathbf{s} = \frac{1}{h \times w} \sum_{i=1}^{h} \sum_{j=1}^{w} \mathbf{F}(i, j). \tag{4}$$

The output $\mathbf{s}$ denotes the global descriptors of the spectral band with the nonlocal statistics of the entire band image. Furthermore, in order to capture bandwise dependences, we employ two-layer 1-D convolution in the spectral dimension. In this way, the bandwise information can be effectively aggregated. More importantly, we utilize the shrinking operation for better learning a nonlinear interaction between spectral bands. As illustrated in Fig. 3, the number of bands is firstly shrieked to $l/a(a = 4)$, then excited back to l. In this way, a nonmutually exclusive relationship could be learned to consider multiple bands rather than one band.

At last, a sigmoidal activation σ is used to normalize the output of the two-layers 1-D convolution

$$\hat{\mathbf{s}} = \sigma\left(f^{2}_{\text{conv1D}}\left(\delta\left(f^{1}_{\text{conv1D}}(\mathbf{s})\right)\right)\right) \in \mathbb{R}^{c*l}. \tag{5}$$

After normalization, the descriptor $\hat{\mathbf{s}}$ could be treated as a band weight, which can be used to selectively emphasize informative band features and suppress less useful ones. The final output of the band attention residual block can be reconstructed by the following formulation:

$$\mathbf{B}_{\text{res}} = \mathbf{X} + \mathbf{F} \otimes \hat{\mathbf{s}} \in \mathbb{R}^{c \times l \times h \times w} \tag{6}$$

where $\mathbf{B}_{\text{res}}$ is the final out of the band attention residual block, and $\hat{\mathbf{s}}$ is the normalized band attention weight. For clarity of expression, $\otimes$ here is dot production with a broadcast mechanism.

C. Loss Function

Design of the perceptual loss function is the most significant for the performance enhancement of the proposed framework.

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To pursue superior spatial enhancement with no spectrum deformation based on better visual impression, the perception loss with spatial loss l_S, spectral loss l_E, and adversarial loss l_A is designed as

$$l_{\text{loss}} = \alpha_1 l_S + \alpha_2 l_E + \alpha_3 l_A. \tag{7}$$

1) Spatial Loss: The spatial loss function as the weighted sum of a content loss l_{l1_loss}, and a proposed weighted TV loss $l_{\text{tv_weight_loss}}$ is

$$l_S = l_{l1_\text{loss}} + \beta_1 l_{\text{tv_weight_loss}} \tag{8}$$

where β_1 is equal to 1.0e−6, l_{1_loss} denotes the l_1 norm between the ground truth $\mathbf{I}^{\text{HR}}$ and the super-resolved image $\mathbf{I}^{\text{SR}}$ calculated as

$$l_{l1_\text{loss}} = \frac{1}{lr^2wh} \sum_{i=1}^{rw} \sum_{j=1}^{rh} \sum_{k=1}^{l} \left| \mathbf{I}^{\text{HR}}_{i,j,k} - G(\mathbf{I}^{\text{LR}})_{i,j,k} \right|_1. \tag{9}$$

In recent years, the pixelwise l_{l1_loss} loss has been a widely employed optimization target in image SR field for its advantages in encouraging less blurring than l_{l2_loss} [42], [43]. To enforce spatial smoothness of the created HSI, a weighted TV loss is proposed by the relationship between neighboring pixels in HR images. For a pixel, the influence of its neighboring pixels in the representation of spatial constraint is unequal, which means that if the respective observed spectra are similar in the HR image, then these pixels are still similar in the reconstructed one. The second formulation of the proposed weighted TV loss is

$$\begin{aligned} l_{\text{tv_weight_loss}} &= \frac{1}{l(rw-1)(rh-1)} \\ &\times \sum_{i=1}^{rw} \sum_{j=1}^{rh} \sum_{k=1}^{l} w_{i,j+1,k} (G(\mathbf{I}^{\text{LR}})_{i,j+1,k} - G(\mathbf{I}^{\text{LR}})_{i,j,k})^2 \\ &+ w_{i+1,j,k} (G(\mathbf{I}^{\text{LR}})_{i+1,j,k} - G(\mathbf{I}^{\text{LR}})_{i,j,k})^2 \end{aligned} \tag{10}$$

where $w_{i,j+1,k} = e^{-|\mathbf{I}^{\text{HR}}_{i,j+1,k} - G(\mathbf{I}^{\text{LR}})_{i,j,k}|}$. The proposed weighted TV loss is beneficial to remove salt-and-pepper noise.

2) Spectral Loss: Spectral information is significant to hyperspectral SR. To avoid spectral distortion, the spectral loss is designed by minimizing the spectral angle between the HR spectral and super-resolved spectral signatures as

$$l_E = \frac{1}{r^2wh} \sum_{i=1}^{rw} \sum_{j=1}^{rh} \arccos \left(\frac{\langle \mathbf{z}_{i,j}, \mathbf{z}'_{i,j} \rangle}{\|\mathbf{z}_{i,j}\|_2 \|\mathbf{z}'_{i,j}\|_2} \right) \tag{11}$$

where $\mathbf{z}_{i,j}$ denotes the spectral vector at the ith row and jth column in HR HSI that is the referenced ground truth. Meanwhile, $\mathbf{z}_{i,j}{}'$ is the reconstructed spectral vector with the same spatial position after SR.

3) Adversarial Loss: The proposed SR framework uses adversarial learning to learn a better generator which can create a super-resolved HSIs I^{SR} that is difficult to be distinguished from ground truth HSI I^{HR} with binary coding (1 for ground truth HSI, 0 for SR HSI). Therefore, similar to the work of [36], after exp(·) function, the adversarial loss is defined as

$$l_A = \frac{1}{N} \sum_{n=1}^{N} (1 + D(G(\mathbf{I}^{\text{LR}}))) \tag{12}$$

where N denotes the number of training samples.

In the proposed SR framework, the spatial constraint plays a significant role in encouraging less blurring for image restoration, the spectral constraint can reduce spectral loss and the adversarial constraint guides generators to generate a more realistic SR result. In our experimental setting, α_1, α_2 and α_3 are empirically set to 1, 1.0e−2, and 1.0e−3.

IV. Experiments

In this section, we conduct a series of experiments on three typical hyperspectral sets to evaluate the performance of our proposed framework. Five methods are selected as the baseline for comparison purposes: Bicubic, SRCNN [9], VDSR [44], LapSRN [45], 3DFCN [29], and GDRRN [46]. Among them, Bicubic is a traditional method; LapSRN, 3DFCN, and GDRRN are the state-of-the-art deep learning-based HSI SR methods; and SRCNN and VDSR are representative deep learning-based algorithms for RGB image SR. In order to make SRCNN and VDSR networks suitable for hyperspectral SR, a three-bandwise manner is employed [29].

A. Data Sets

Different types of HSIs are applied for evaluation. Data sets from well-known hyperspectral sensors, namely ROSIS, are real data, and CAVE database is a synthesized one. Pavia University and Pavia Center are two hyperspectral scenes acquired via the ROSIS sensor during a flight campaign over Pavia, northern Italy. The ROSIS sensor covers 115 spectral bands from 0.43 to 0.86 μm. The geometric resolution is 1.3 m. The University of Pavia image has 610 × 340 pixels, each having 103 bands after bad-band and noisy band removal. The Pavia Center scene contains 1096 × 1096 pixels, 102 spectral bands left after bad-band removal. In the Pavia Center scene, only 1096 × 715 valid pixels are utilized after discarding the samples with no information in this experiment. The CAVE database is used to further evaluate the proposed framework. It includes 32 indoor images showing, for instance, paintings, toys, and food, captured under controlled illumination. Their wavelengths cover the visible spectrum from 400 to 700 nm with a 10-nm interval band. The size of all these HSIs in CAVE is 512 × 512 with 31 spectral bands.

B. Evaluation Criterion

To comprehensively evaluate the performance of the proposed methods, four classical evaluation metrics are utilized: mean peak signal-to-noise ratio (MPSNR), mean structural similarity index (MSSIM), mean root-mean-square error (MRMSE), and spectral angle mapper (SAM). The PSNR (in dB), SSIM, and MRMSE are three traditional image quality measures. They compare the reconstructed high spatial-resolution image with the ground-truth image in terms of

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TABLE I
QUANTITATIVE COMPARISON OF HYPERSPECTRAL SR OF THE PAVIA UNIVERSITY DATA SET

	SNR(db)		Bicubic	SRCNN	VDSR	LapSRN	3DFCN	GDRRN	Proposed
Pavia university	∞	MPSNR	26.968	27.688	28.056	28.119	28.614	27.817	**30.612**
		MSSIM	0.7063	0.7552	0.7728	0.7716	0.7885	0.7798	**0.8730**
		MRMSE	11.741	10.798	10.345	10.306	9.7814	10.607	**7.8977**
		SAM	5.7950	6.0594	5.8123	5.1776	5.1709	6.5059	**4.1083**
	60	MPSNR	26.966	27.687	28.054	28.117	28.612	27.817	**30.612**
		MSSIM	0.7063	0.7552	0.7726	0.7715	0.7884	0.7798	**0.8730**
		MRMSE	11.743	10.801	10.347	10.309	9.7829	10.621	**7.8984**
		SAM	5.8004	6.0604	5.8213	5.1866	5.1771	6.5149	**4.1112**
	30	MPSNR	26.941	27.662	28.042	28.012	28.598	27.805	**30.601**
		MSSIM	0.7052	0.7540	0.7706	0.7662	0.7842	0.7789	**0.8719**
		MRMSE	11.845	10.811	10.459	10.475	9.8079	10.620	**7.8991**
		SAM	5.8113	6.0836	5.8391	5.5124	5.3955	6.5165	**4.1120**

spatial quality of each band. The MPSNR and MRMSE can estimate the similarity between the generated image and the ground-truth image by the mean squared error. The MSSIM emphasizes the structural consistency with the ground-truth image. The SAM is utilized to evaluate the spectral reconstruction quality of each spectrum at pixel-levels, which calculate the average angle between spectral vectors of the generated HSI and the ground-truth one. Given a reconstruction image $\hat{\mathbf{P}}$ and a ground truth image $\mathbf{P}$, these four evaluation criteria can be defined as follows:

$$\text{MPSNR} = \frac{1}{l}\sum_{k=1}^{l} 10 \times \lg\left[\frac{(\text{MAX}_k)^2}{\frac{1}{wh}\sum_{i=1}^{w}\sum_{j=1}^{h}\left(\hat{\mathbf{P}}(i,j)-\mathbf{P}(i,j)\right)^2}\right] \tag{13}$$

$$\text{MSSIM} = \frac{1}{l}\sum_{k=1}^{l}\frac{(2\mu_{\hat{\mathbf{P}}}\mu_{\mathbf{P}}+C_1)(2\sigma_{\hat{\mathbf{P}}P}+C_2)}{(\mu_{\hat{\mathbf{P}}}^2+\mu_{\mathbf{P}}^2+C_1)(\sigma_{\hat{\mathbf{P}}}^2+\sigma_{\mathbf{P}}^2+C_2)} \tag{14}$$

$$\text{MRMSE} = \frac{1}{l}\sum_{k=1}^{l}\sqrt{\frac{1}{wh}\sum_{i=1}^{w}\sum_{j=1}^{h}\left(\hat{\mathbf{P}}(i,j)-\mathbf{P}(i,j)\right)^2} \tag{15}$$

$$\text{SAM} = \frac{1}{n}\sum \arccos\left(\frac{\langle \mathbf{z_i}, \mathbf{z_i}'\rangle}{\|\mathbf{z_i}\|_2\|\mathbf{z_i}'\|_2}\right) \tag{16}$$

where MAX_k is the maximum intensity in the kth band, k denotes the index of the band, $\mu_{\hat{\mathbf{P}}}$ and $\mu_{\mathbf{P}}$ are the mean values of $\hat{\mathbf{P}}$ and $\mathbf{P}$, respectively, $\sigma_{\hat{\mathbf{P}}}$ and $\sigma_{\mathbf{P}}$ are the variances of $\hat{\mathbf{P}}$ and $\mathbf{P}$, respectively, $\sigma_{\hat{\mathbf{P}}P}$ is the covariance between $\hat{\mathbf{P}}$ and $\mathbf{P}$, C_1 and C_2 are two constants used to avoid instability, $n = w \times l$ is the number of pixels, $\langle \mathbf{z_i}, \mathbf{z_i}'\rangle$ denotes the dot product of two spectra $\mathbf{z_i}$ and $\mathbf{z_i}'$, and $\|\cdot\|_2$ represents the l_2 norm operation.

In these evaluation metrics, the larger the MPSNR and MSSIM, the similar the reconstructed HSI and the ground truth one. Meanwhile, the more smaller the SAM and MRMSE, more similar the reconstructed and the ground-truth ones.

C. Parameter Setting

For the experiments on the Pavia University data sets, a 144 $\times$ 144 subregion is selected to evaluate the performance of our proposed method, while the remaining are used for training. For the CAVE data set, we randomly select 20 images as the training data set, the remaining 12 images are utilized as the testing data set. To synthesize the LR samples, an HR patch is blurred by a Gaussian kernel and sub-sampled by the upscaling factor. All the LR patches are obtained by downsampling the HR patches with a factor of 2, 4, and 8, and the LR patches are of spatial size 72 $\times$ 72, 36 $\times$ 36, and 18 $\times$ 18, respectively. For each upscaling factor belonging to 2, 4, 8, a specific network is trained for the corresponding factor. In the test processing, the generator model is applied on the images of arbitrary sizes. The proposed SR framework is implemented on the fast-deep learning framework named Pytorch. Training was performed on an NVIDIA GTX1080Ti GPU. The parameters of the network were optimized using Adam modification of stochastic gradient descent with $\beta = 0.9$ and the learning rate is set to $1.0e-3$. The results obtained from Pavia University data are displayed by the 49th band, the results obtained from Pavia Center data are displayed by the 13th band, and the results obtained from cave data are displayed by the 28th band.

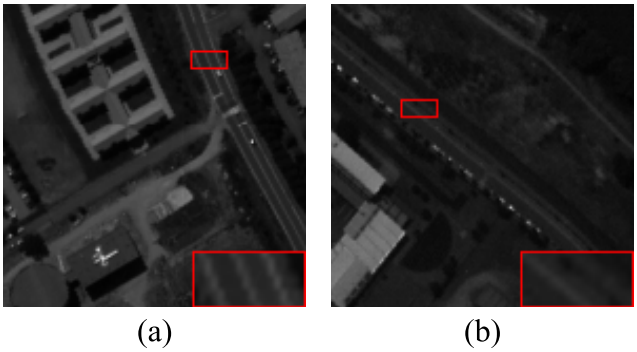


Fig. 4. Example groundtruth of HR HSIs. (a) Pavia University data set. (b) Pavia Center data set.

D. Experimental Results on Real-Scenario HSIs Acquired by ROSIS Sensor

The performance of our proposed SR algorithm is estimated via the two frequently utilized the data sets acquired from

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TABLE II
QUANTITATIVE COMPARISON OF HYPERSPECTRAL SR OF THE PAVIA CENTER DATA SET

	SNR(db)		Bicubic	SRCNN	VDSR	LapSRN	3DFCN	GDRRN	Proposed
Pavia Center	∞	MPSNR	28.596	29.066	29.559	29.778	30.337	29.496	**31.493**
		MSSIM	0.7172	0.7623	0.7826	0.7786	0.8008	0.7723	**0.8508**
		MRMSE	10.245	9.5795	9.0667	8.9334	8.4144	9.1767	**7.7231**
		SAM	6.7111	7.5744	6.9046	6.3491	6.3486	6.5956	**5.3694**
	60	MPSNR	28.593	29.062	29.556	29.776	30.336	29.494	**31.491**
		MSSIM	0.7172	0.7623	0.7827	0.7784	0.7998	0.7722	**0.8507**
		MRMSE	10.246	9.5805	9.0689	8.9360	8.4137	9.1779	**7.7239**
		SAM	6.7146	7.5844	6.9087	6.3561	6.3516	6.5994	**5.3638**
	30	MPSNR	28.394	28.960	29.445	29.671	30.225	29.181	**31.425**
		MSSIM	0.7091	0.7601	0.7721	0.7704	0.7894	0.7573	**0.8455**
		MRMSE	10.348	9.5939	9.0767	8.9902	8.5935	9.4169	**7.7405**
		SAM	6.7347	7.5940	6.9262	6.5774	6.5250	7.5433	**5.4064**

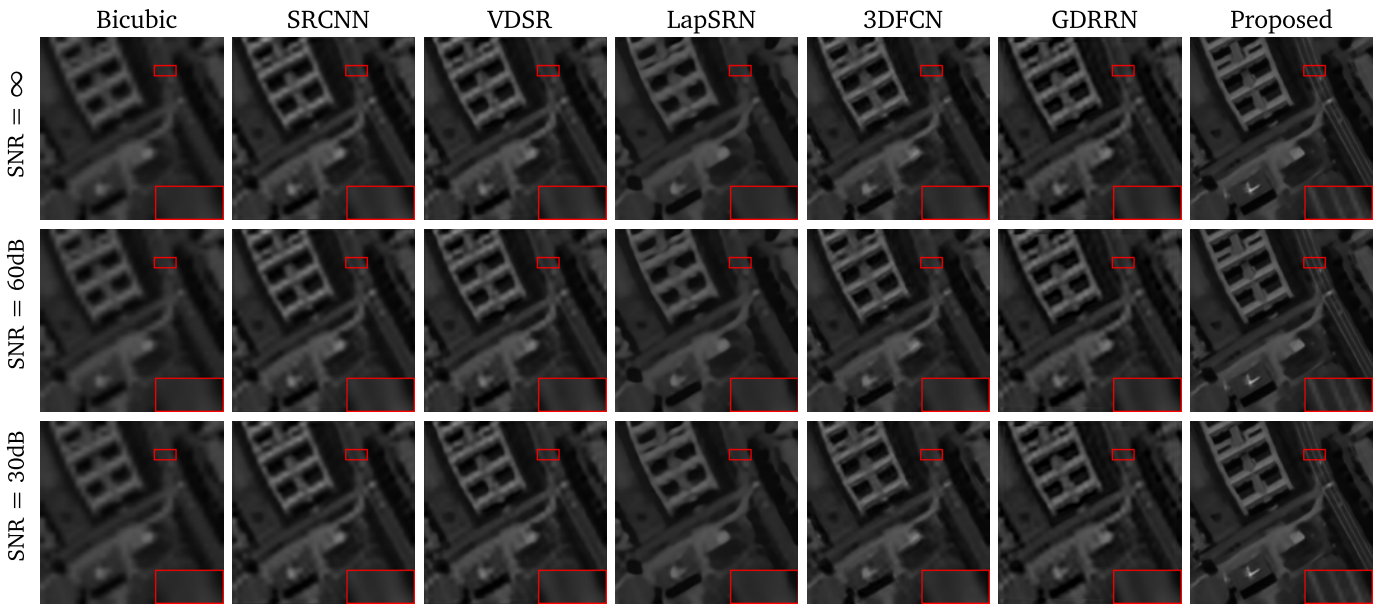


Fig. 5. Visual results of different SR methods on the Pavia University data set, and the area in the red rectangle is enlarged three times in the bottom right corner of the image.

ROSIS sensor, namely the Pavia University data set and the Pavia Center data set.

1) SR Results Under Different SNRs: We first experimented with a 4× sampling factor to evaluate the proposed method. In addition, in order to verify the robustness of the proposed algorithm to noise, three different levels with additive Gaussian white noise, computed via SNR, are added to the LR HSIs in the testing process. Fig. 4 depicts the related HR groundtruth. Tables I and II tabulate the quantitative comparison results of the Pavia University data set and Pavia Center data set under different levels of Gaussian white noise, respectively. Furthermore, their corresponding visual results are demonstrated in Figs. 5 and 6. For the results achieved from the Pavia University data set, the proposed method shows the following.

1) The proposed method significantly outperforms other comparative ones with the highest MPSNR and MSSIM and lowest MRMSE and SAM values for all the cases.
2) Compared with the Bicubic algorithm, the improvements of our methods are 3.644 dB and 0.1667 in terms of MPSNR and MSSIM; meanwhile, the decreases are 3.8433 and 1.6867 in terms of MRMSE and SAM. Furthermore, the proposed algorithm still works better under different levels of Gaussian white noise. MPSNR increased by an average of 3.653 dB and SAM dropped by an average of 1.6943 when SNR under 60 and 30 dB.
3) The proposed method achieves the best spatial reconstruction than other SR methods irrelevant to the level of SNR. It can also be clearly seen from Fig. 5 that the image reconstructed by the proposed method has finer texture details than other comparison methods.
4) The proposed method achieves the best spectral fidelity with the lowest SAM in all the cases. Fig. 7 further demonstrates the spectrum after hyperspectral SR. It is implied that the spectra reconstructed by the proposed

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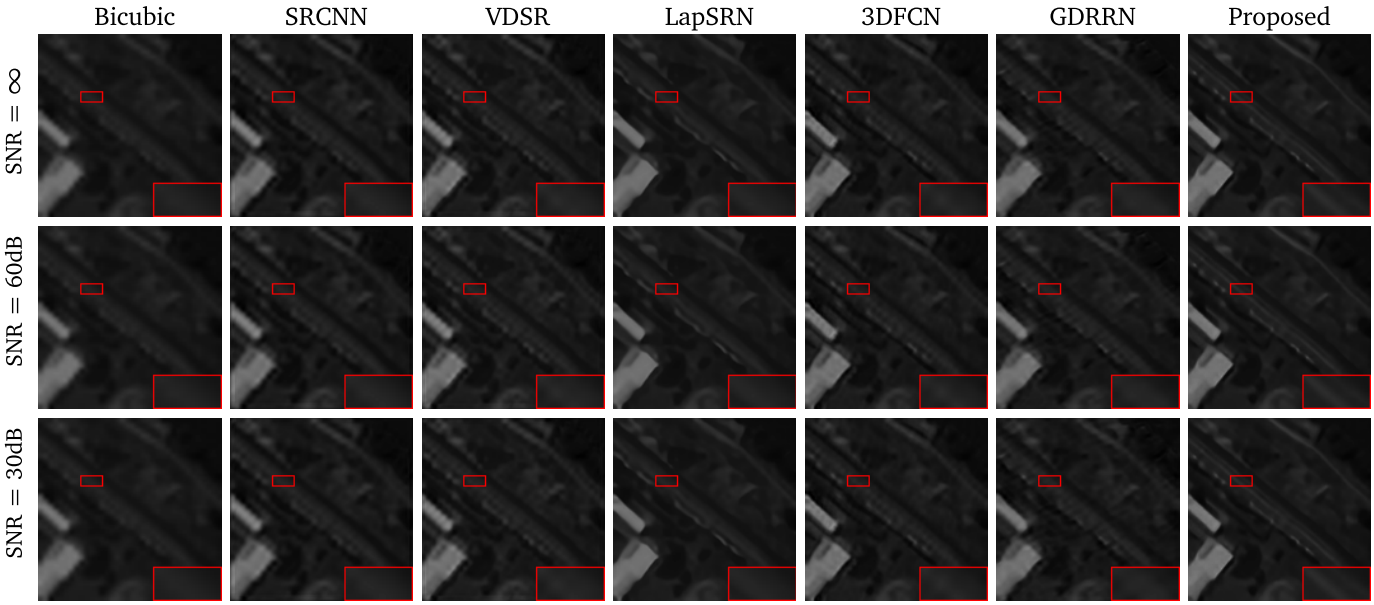


Fig. 6. Visual results of different SR methods on the Pavia Center data set, and the area in the red rectangle is enlarged three times in the bottom right corner of the image.

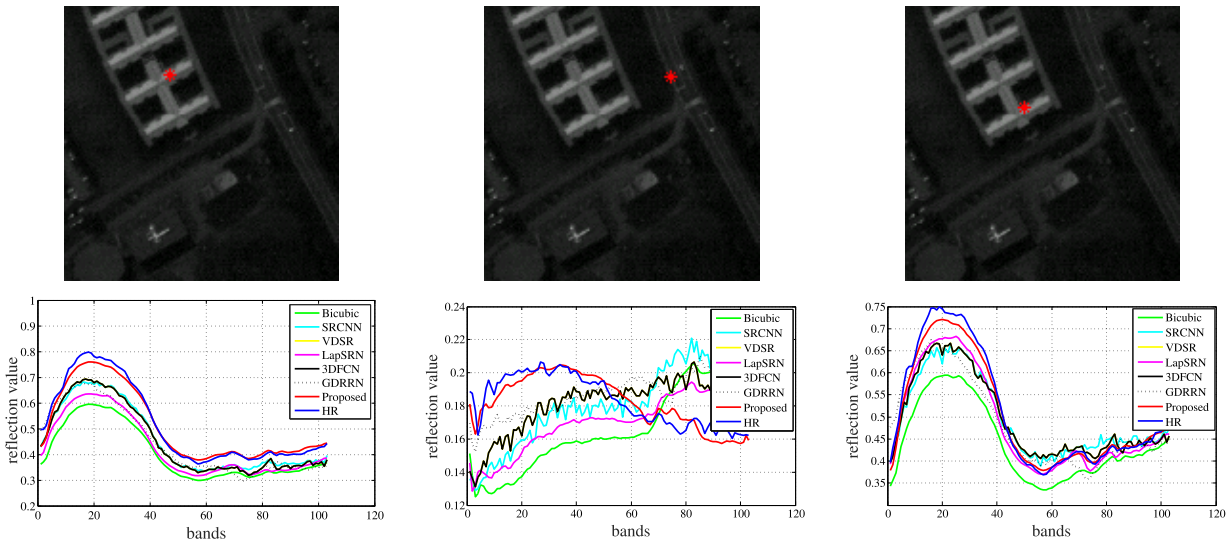


Fig. 7. Example spectrum of Pavia University data set. (Top) Locations. (Bottom) Corresponding spectrum.

algorithm is closer to its groundtruth than those achieved by comparison methods.

From the results obtained on the Pavia Center data set in Fig. 6, we can see that compared with other methods, our proposed HR method has superior HR results. For instance, compared with the Bicubic algorithm, MPSNR increased by an average of 2.942 dB, MSSIM improved by 0.1345, MRMSE decreased by 2.5505, and SAM dropped of 1.3403 when SNR under ∞, 60, and 30 dB. Fig. 8 depicts the spectral accuracy achieved via different SR methods, and it indicates that our proposed SR method can reconstruct more similar spectra than other methods. Therefore, the results on the Pavia Center data set have similar phenomena as the result on the Pavia University data set. From the aforementioned results, it can be concluded that the proposed algorithm obviously outperforms other algorithms on these two data sets acquired by the ROSIS sensor.

2) SR Results Using Different Up-Sampling Factors: In this section, we design a series of experiments to discuss SR results with different upsampling factors. Tables III and IV tabulate the quantitative results by using different SR methods with upsampling factors 2, 4, and 8, over the two Pavia data sets. Their corresponding visual results are demonstrated in Figs. 9 and 10. Our proposed method performs favorably against other comparison algorithms with different upsampling factors. Despite the difficulty of SR is at large upsampling factor, our $8\times$ model still performs well. For the Pavia University data set, compared with the Bicubic method, the proposed algorithm achieves an improvement of 3.998 dB in MPSNR, and 1.8261 in SAM. From the results obtained over the Pavia

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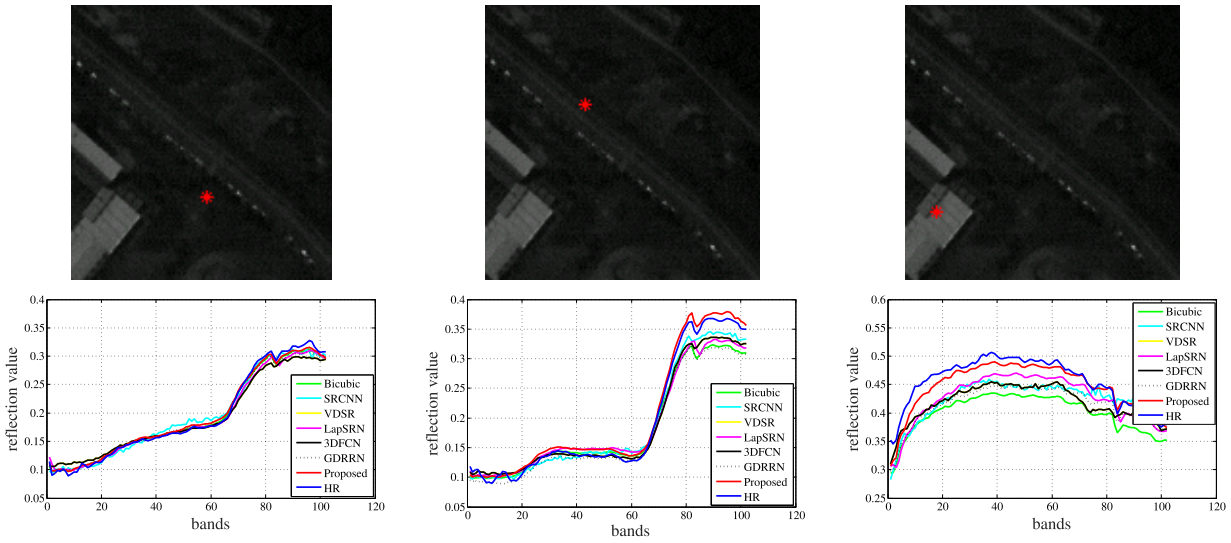


Fig. 8. Example spectrum of Pavia Center data set. (Top) Locations. (Bottom) Corresponding spectrum.

TABLE III

QUANTITATIVE COMPARISON OF HYPERSPECTRAL SR VIA DIFFERENT UP-SAMPLING FACTORS OF THE PAVIA UNIVERSITY DATA SET

	up-sampling		Bicubic	SRCNN	VDSR	LapSRN	3DFCN	GDRRN	Proposed
Pavia university	2	MPSNR	30.899	32.165	33.130	33.202	34.499	32.484	**36.619**
		MSSIM	0.8824	0.9174	0.9316	0.9311	0.9430	0.9225	**0.9699**
		MRMSE	7.4590	6.4045	5.7348	5.7363	5.0285	6.1801	**3.8990**
		SAM	3.9093	4.5475	4.1408	3.283	3.2572	4.3918	**2.7665**
	4	MPSNR	26.968	27.688	28.056	28.119	28.614	27.817	**30.612**
		MSSIM	0.7063	0.7552	0.7728	0.7716	0.7885	0.7798	**0.8730**
		MRMSE	11.741	10.798	10.345	10.306	9.7814	10.607	**7.8977**
		SAM	5.7950	6.0594	5.8123	5.1776	5.1709	6.5059	**4.1083**
	8	MPSNR	24.207	24.676	24.902	24.942	25.233	24.770	**26.837**
		MSSIM	0.5391	0.5705	0.5824	0.5846	0.6008	0.5767	**0.7028**
		MRMSE	16.140	15.2701	14.883	14.810	14.387	15.107	**12.407**
		SAM	8.2677	8.1907	7.9874	7.5782	7.6092	8.2465	**5.6190**

TABLE IV

QUANTITATIVE COMPARISON OF SPATIAL SR FOR DIFFERENT UP-SAMPLING FACTORS OVER THE PAVIA CENTER DATA SET

	up-sampling		Bicubic	SRCNN	VDSR	LapSRN	3DFCN	GDRRN	Proposed
Pavia Center	2	MPSNR	32.725	33.910	35.166	35.165	36.195	34.345	**37.340**
		MSSIM	0.8895	0.9264	0.9371	0.9354	0.9460	0.9241	**0.9647**
		MRMSE	6.3322	6.017	5.0566	4.7600	4.2367	5.1889	**3.8140**
		SAM	4.9766	5.347	4.7103	4.3839	4.3234	4.9905	**3.7758**
	4	MPSNR	28.596	29.066	29.559	29.778	30.337	29.496	**31.493**
		MSSIM	0.7172	0.7623	0.7826	0.7786	0.8008	0.7723	**0.8508**
		MRMSE	10.245	9.5795	9.0667	8.9334	8.4144	9.1767	**7.7231**
		SAM	6.7111	7.5744	6.9046	6.3491	6.3486	6.5956	**5.3694**
	8	MPSNR	25.889	26.171	26.318	26.378	26.734	26.309	**27.984**
		MSSIM	0.5578	0.5879	0.5987	0.5839	0.6073	0.5985	**0.6703**
		MRMSE	14.029	13.492	13.262	13.297	12.838	13.315	**12.043**
		SAM	8.5169	8.8558	8.6149	8.4872	8.2453	8.2204	**7.1990**

Center data set, it can be seen that the MPSNR of our proposed algorithm is 4.615, 2.897, and 2.095 dB larger than the MPSNR obtained from GDRRN with 2×, 4× and 8× upsampling factors, respectively. Obviously, our 8× model provides a competitive performance to other state-of-the-art methods.

3) Fine-Tune Results Between Two Data Sets Acquired by ROSIS Sensor: Considering that the Pavia data sets are

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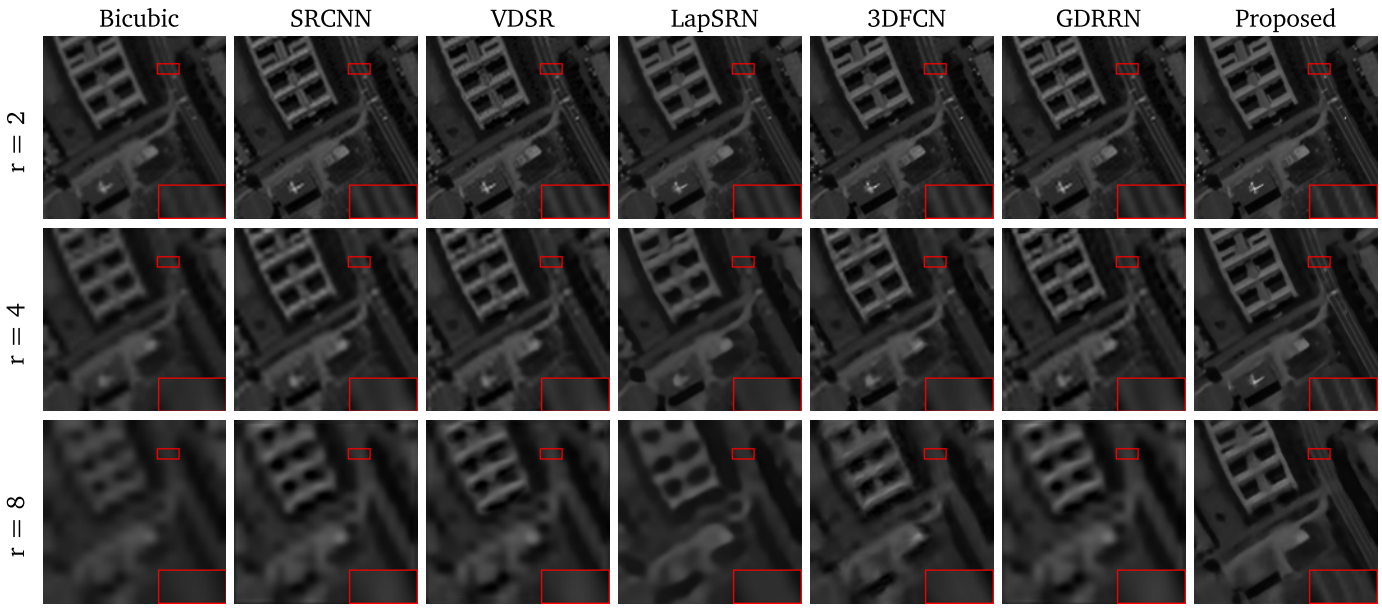


Fig. 9. Visual results of hyperspectral SR via different up-sampling factors on the Pavia University data set.

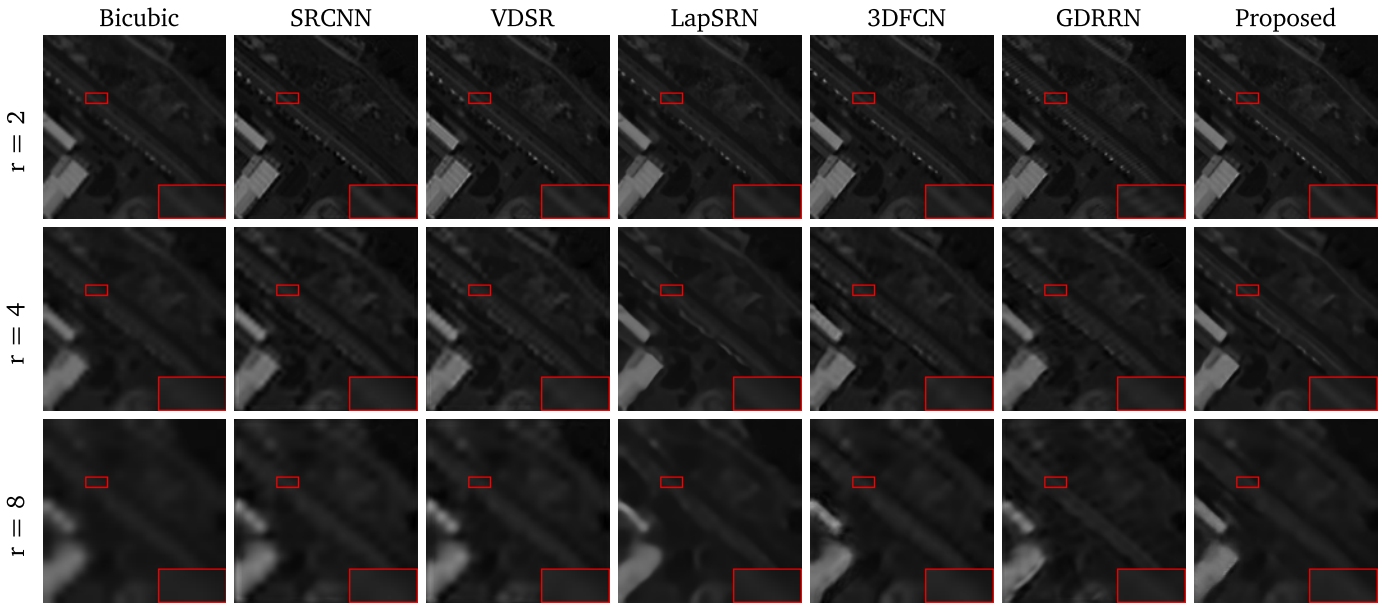


Fig. 10. Visual results of spatial SR for different up-sampling factors on the Pavia Center data set.

TABLE V
QUANTITATIVE COMPARISON OF SPATIAL SR FOR FINE-TUNE OVER THE PAVIA UNIVERSITY DATA SET BASED ON THE PAVIA CENTER MODEL

up-sampling		without fine tune	finetune			
			00%	60%	80%	100%
Pavia University	4	MPSNR	30.612	30.041	30.605	30.706	**30.853**
		MSSIM	0.8730	0.8581	0.8746	0.8758	**0.8801**
		MRMSE	7.8977	8.3027	7.9020	7.8115	**7.6670**
		SAM	4.1083	4.3521	4.1569	4.0818	**4.0329**

acquired from the same sensor, we fine-tune them with each other. Specifically, we employed the model trained on the Pavia Center data set as the initial for SR of the Pavia University data set, and the model trained on the Pavia University data set as the initial for spatial SR of the Pavia Center data set. In addition, parameters of the proposed framework are

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TABLE VI
QUANTITATIVE COMPARISON OF SPATIAL SR FOR FINE-TUNE OVER THE PAVIA CENTER DATA SET BASED ON THE PAVIA UNIVERSITY MODEL

up-sampling		without fine tune	finetune			
			00%	60%	80%	100%
Pavia Center	4	MPSNR	31.493	30.481	31.404	31.418	**31.564**
		MSSIM	0.8508	0.8154	0.8462	0.8463	**0.8521**
		MRMSE	7.7231	8.3204	7.8076	7.8116	**7.6674**
		SAM	5.3694	6.2662	5.3762	5.3723	**5.3253**

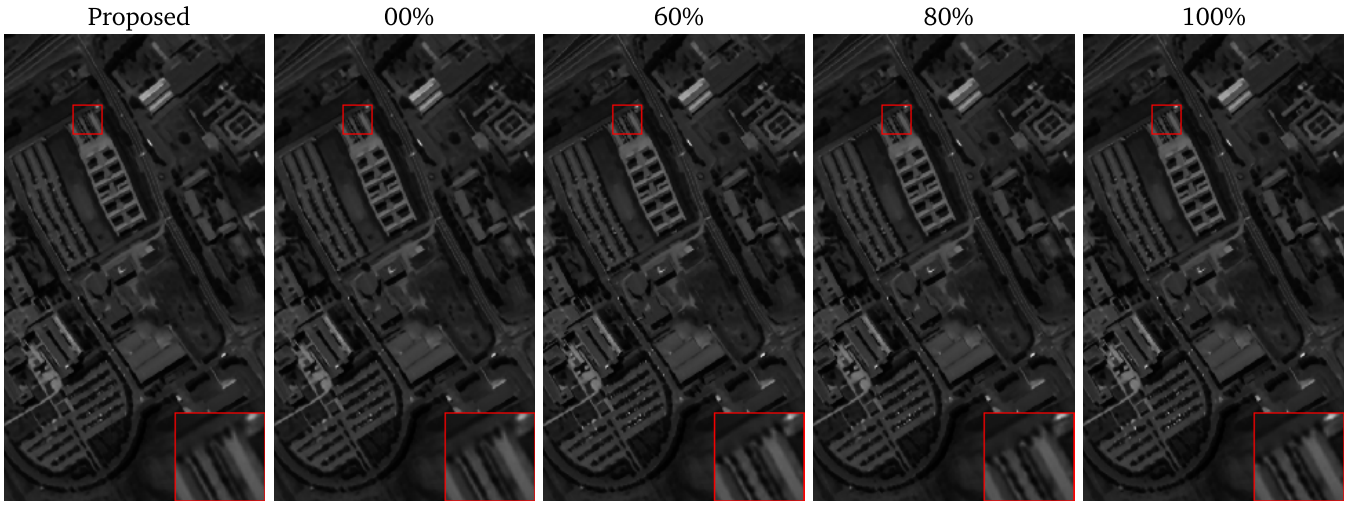


Fig. 11. Visual results of spatial SR for fine-tune over the Pavia University data set based on the Pavia Center model.

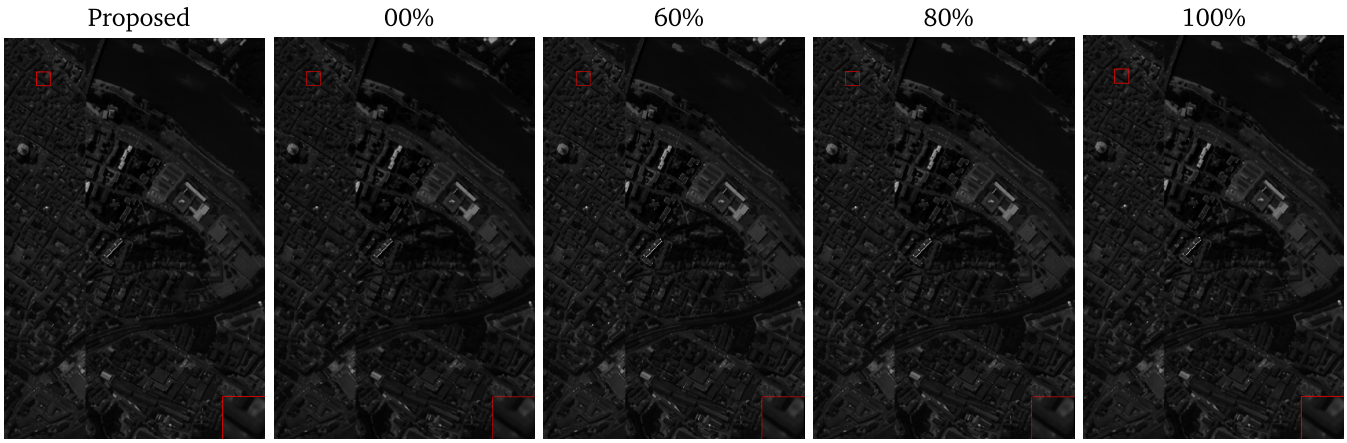


Fig. 12. Visual results of spatial SR for fine-tune over the Pavia Center data set based on the Pavia University model.

fine-tuned with a limited number of available training samples from the target data sets. In order to test the performance of fine-tune for SR, we randomly selected 60%, 80%, and 100% of the original training samples as the training samples, and kept the testing samples the same size as before. Besides, the results without employing the original target data noted as 00% are also tested and tabulated. Tables V and VI list quantitative results from our proposed algorithm of the Pavia University and Pavia Center data sets, respectively. Without fine-tune denotes that the results achieved by using the samples from the target scene and the trained model have not been fine-tuned. Their corresponding visual results at full resolution are shown in Figs. 11 and 12. For the Pavia University data set, it is observed that, when testing the Pavia University data set directly on a well-trained model of the Pavia Center data set without using any data of Pavia University, our model performs well with the improvements of 3.073 dB in MPSNR and decrease of 1.363 in SAM compared with the Bicubic method. With the increasing original data, the SR performance becomes superior, especially when utilizing 80% original data, the SR accuracy is better than the accuracy obtained by the proposed algorithm. From Table VI, compared with the 3DFCN algorithm, the MPSNR obtained by without using the data of Pavia Center increased by 0.144 dB, MSSIM improved

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TABLE VII
QUANTITATIVE COMPARISON OF HYPERSPECTRAL SR OF THE CAVE DATA SET

	SNR(db)		Bicubic	SRCNN	VDSR	LapSRN	3DFCN	GDRRN	Proposed
Cave	∞	MPSNR	34.196	35.399	35.847	35.968	36.958	35.583	**38.350**
		MSSIM	0.8942	0.9090	0.9145	0.9202	0.9243	0.9112	**0.9469**
		MRMSE	6.7406	5.7973	5.4982	5.4043	5.0379	5.6602	**4.3579**
		SAM	4.2567	4.6055	4.4642	4.9059	4.1944	4.5794	**3.6786**
	60	MPSNR	34.190	35.394	35.842	35.948	36.940	35.578	**38.341**
		MSSIM	0.8941	0.9089	0.9145	0.9200	0.9241	0.9112	**0.9467**
		MRMSE	6.7430	5.7977	5.4987	5.4112	5.0463	5.6608	**4.3609**
		SAM	4.2658	4.6525	4.5127	4.9207	4.2671	4.6273	**3.6831**
	30	MPSNR	33.173	34.421	34.793	34.891	35.889	34.577	**37.714**
		MSSIM	0.8939	0.8910	0.8960	0.9194	0.9236	0.8929	**0.9367**
		MRMSE	6.9498	6.0374	5.7534	5.7314	5.5591	5.9073	**4.5686**
		SAM	8.2907	8.1215	8.0517	7.9611	7.2427	8.1309	**5.6960**

TABLE VIII
QUANTITATIVE COMPARISON OF HYPERSPECTRAL SR FOR DIFFERENT UP-SAMPLING FACTORS OVER THE CAVE DATA SET

	up-sampling		Bicubic	SRCNN	VDSR	LapSRN	3DFCN	GDRRN	Proposed
Cave	2	MPSNR	39.270	41.600	41.895	42.049	43.472	41.771	**44.663**
		MSSIM	0.9595	0.9704	0.9716	0.9738	0.9766	0.9711	**0.9849**
		MRMSE	3.7817	2.8195	2.7198	2.6865	2.3352	2.7606	**2.2040**
		SAM	3.2369	3.1500	3.0884	3.3596	3.0065	3.1224	**2.7059**
	4	MPSNR	34.196	35.399	35.847	35.968	36.958	35.583	**38.350**
		MSSIM	0.8942	0.9090	0.9145	0.9202	0.9243	0.9112	**0.9469**
		MRMSE	6.7406	5.7973	5.4982	5.4043	5.0379	5.6602	**4.3579**
		SAM	4.2567	4.6055	4.4642	4.9059	4.1944	4.5794	**3.6786**
	8	MPSNR	30.441	30.671	31.101	31.345	32.009	30.751	**33.431**
		MSSIM	0.8152	0.8203	0.8287	0.8257	0.8417	0.8217	**0.8871**
		MRMSE	10.227	9.6365	9.3074	8.8562	8.7251	9.6205	**7.2917**
		SAM	5.6215	6.9828	6.3472	6.9150	6.1252	6.7568	**5.0489**

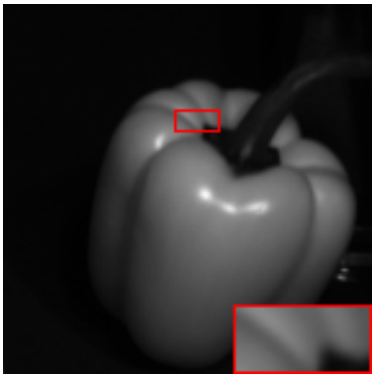


Fig. 13. Example groundtruth (HR) on the CAVE data set.

by 0.0146, MRMSE decreased by 0.094, and SAM dropped of 0.0824. Figs. 11 and 12 and Tables V and VI indicate that our proposed fine-tune for SR is very effective and that avoids the requirement of a huge amount of training samples from the target scene.

E. Experiments on Synthesized HSIs From CAVE Data Set

To further evaluate the performance of our proposed SR algorithm for SR, a set of similar experiments were performed on the synthesized CAVE data set.

1) SR Results Under Different SNRs: We first utilize the $4\times$ sampling factor to evaluate the proposed algorithm method. Similarly, we tested the robustness of the proposed algorithm to noises when different levels of additive Gaussian white noise are added to the LR HSIs. As baselines to compare against, we still use six methods: Bicubic, SRCNN, VDSR, LapSRN, 3DFCN, and GDRRN. Fig. 13 is the related ground truth (HR) image. Table VII depicts the quantitative comparison results of the Cave data set under different levels of Gaussian white noise. Correspondingly, their visual results are shown in Fig. 14. The proposed method shows the following.

1) The proposed framework significantly outperforms other competing ones with the best MPSNR, MSSIM, and MRMSE values for all the cases in the CAVE data set.
2) The proposed framework is robust under different levels of Gaussian white noise. Compared with the Bicubic method, the MPSNR of our algorithm increased by an average of 3.943 dB, the MSSIM improved by an average of 0.0494, the MRMSE dropped by an average of 2.382 and the SAM decreased by an average of 1.2548.
3) In terms of spatial reconstruction, the proposed method produced much sharper edges than other competing methods.

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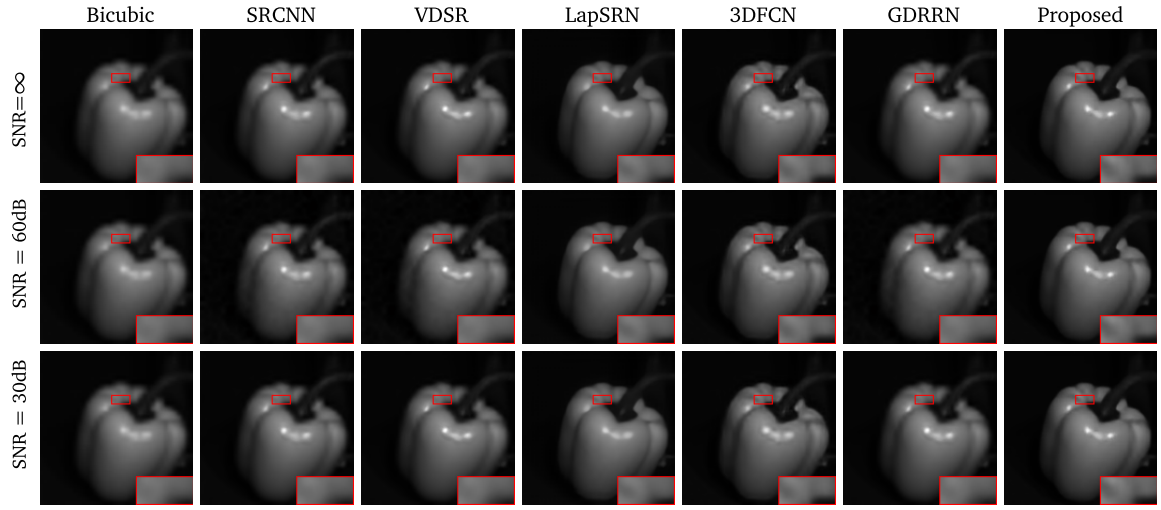


Fig. 14. Visual results of spatial SR on CAVE data set, in which the area in the red rectangle is enlarged three times in the bottom right corner of the image.

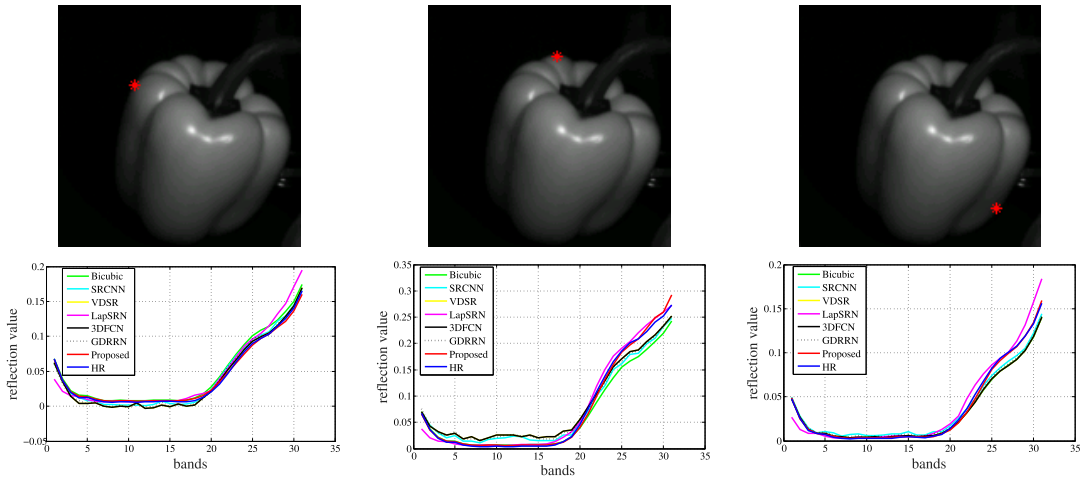


Fig. 15. Example spectrum of CAVE data set. (Top) Locations. (Bottom) Corresponding spectrum.

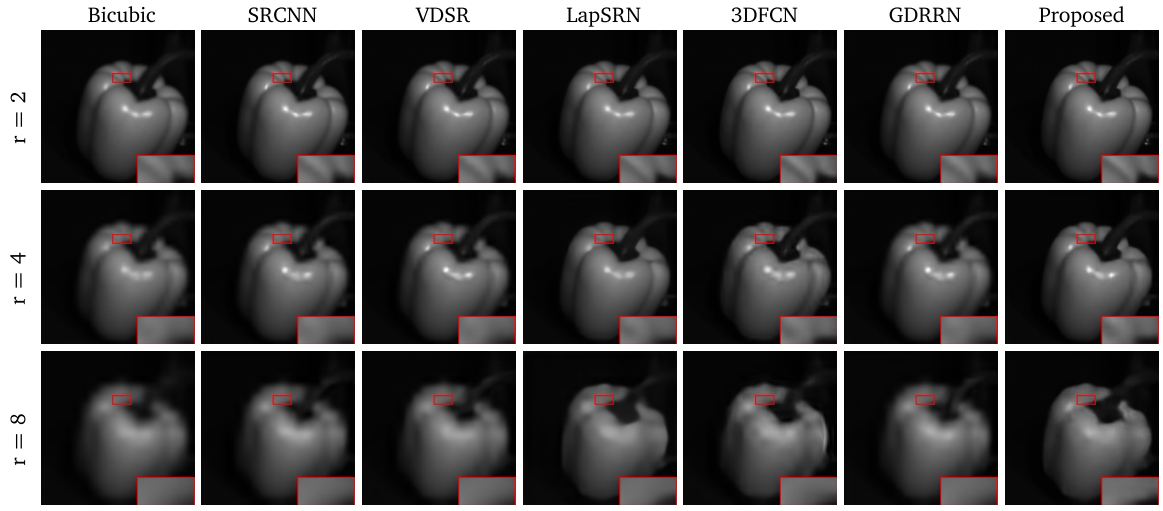


Fig. 16. Visual results of spatial SR for different up-sampling factors on the CAVE data set in which the area in the red rectangle is enlarged three times in the bottom right corner of the image.

Fig. 15 demonstrates the resolved spectra after spatial SR. From Figs. 14 and 15 and Table VII, we can see that our approach not only perform well on SAM, but also achieve visually more appealing results.

2) SR Results Using Different Upsampling Factors: Table VIII tabulates the quantitative results by different SR methods with upsampling factors 2, 4, and 8. Their corresponding visual results are demonstrated in Fig. 16. As shown

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in Table VIII and Fig. 16, our proposed method yields superior performance to other comparison methods with different upsampling factors in terms of MPSNR, MSSIM, and MRMSE. Our model is more capable of handling the SR task at large upsampling factors.

V. Conclusion

In this article, we propose a hyperspectral SR framework with a novel architecture based on band attention through adversarial learning. The generator is designed based on three residual blocks with an identical layout by using 3-D CNNs. Considering the spatial enhancement while avoiding spectral distortion, band attention mechanism is applied and a novel perceptual loss consisting of spatial, spectral loss, and adversarial loss is designed to benefit the HR due to change invariant in pixel level. Experimental results demonstrate that our proposed framework outperforms existing methods, and can super-resolve the LR HSIs with finer texture features. Therefore, it is a recommended framework yielding better results not only in quantitative measurement but also in visual display.

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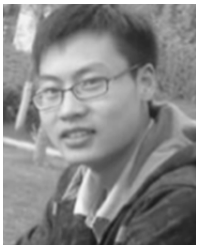
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